

Data Schema & Storage Architecture

AI-Powered Governance Intelligence Platform Specification

1. Canonical PostgreSQL Data Models & Entities

Table 1: **dim_ministries**

Stores central government administrative bodies.

Field Name	Type	Constraints	Relations / Foreign Keys	Description
ministry_id	UUID	PK, Default gen_random_uui... ▾	None ▾	Primary Key
ministry_code	VARCHAR(32)	Unique, Not Null ▾	None ▾	Unique identifier (e.g., GOI_MORD)
ministry_name	TEXT	Not Null ▾	None ▾	Full title (e.g., Ministry of Rural Development)
nodal_email	VARCHAR(255)	Not Null ▾	None ▾	Administrative contact address
created_at	TIMESTAMPTZ	Default clock_timestamp() ▾	None ▾	Record creation audit timestamp

Table 2: **dim_schemes**

Catalog of centrally sponsored and central sector schemes.

Field Name	Type	Constraints	Relations / Foreign Keys	Description
scheme_id	UUID ▾	PK, Default gen_random_uui... ▾	None ▾	Primary Key
ministry_id	UUID ▾	Not Null ▾	FK → dim_ministries.ministry... ▾	Parent ministry reference
scheme_code	VARCHAR(64) ▾	Unique, Not Null ▾	None ▾	Official code (e.g., PMAY_G, MGNREGA)
scheme_name	TEXT ▾	Not Null ▾	None ▾	Full public scheme name
scheme_type	VARCHAR(32) ▾	Check (type IN ('CORE_OF_... ▾	None ▾	Official financial scheme classification
launch_date	DATE ▾	Optional ▾	None ▾	Operational launch date
is_active	BOOLEAN ▾	Default true ▾	None ▾	Active ingestion flag

Table 3: dim_lgd_geography

Canonical spatial and administrative directory conforming to Ministry of Panchayati Raj LGD standards.

Field Name	Type	Constraints	Relations / Foreign Keys	Description
lgd_code	INTEGER	PK	None	Official Government LGD numeric code
entity_level	VARCHAR(16)	Check (entity_level IN ('STAT...	None	Administrative tier
entity_name	VARCHAR(128)	Not Null	None	Canonical English name
parent_lgd	INTEGER	Nullable (Null for States)	FK → dim_lgd_geography.lg...	Self-referencing hierarchical parent
state_lgd	INTEGER	Not Null	None	Associated root state LGD code
geojson	JSONB	Nullable	None	Vector boundary geometry for GIS rendering

Table 4: fact_scheme_allocations

Financial allocations, releases, and reported expenditures aggregated by time and geography.

Field Name	Type	Constraints	Relations / Foreign Keys	Description
allocation_id	UUID	PK, Default gen_random_uui...	None	Primary Key
scheme_id	UUID	Not Null	FK → dim_schemes.scheme...	Target scheme reference
lgd_code	INTEGER	Not Null	FK → dim_lgd_geography.lg...	Geographic allocation destination
financial_year	VARCHAR(9)	Not Null (e.g., '2025-2026')	None	Indian Financial Year period
quarter	SMALLINT	Check (quarter BETWEEN 1 ...	None	Reporting fiscal quarter
allocated_cr	NUMERIC(14,2)	Not Null, Default 0.00	None	Budget allocation in ₹ Crores
released_cr	NUMERIC(14,2)	Not Null, Default 0.00	None	Central treasury release in ₹ Crores
utilized_cr	NUMERIC(14,2)	Not Null, Default 0.00	None	Validated expenditure in ₹ Crores
updated_at	TIMESTAMPZ	Default clock_timestamp()	None	Timestamp of last reported progress

Table 5: fact_scheme_beneficiaries_agg

De-duplicated, aggregated beneficiary and physical milestone footprints.

Field Name	Type	Constraints	Relations / Foreign Keys	Description
metric_id	UUID ▾	PK, Default gen_random_uui... ▾	None ▾	Primary Key
scheme_id	UUID ▾	Not Null ▾	FK → dim_schemes.scheme... ▾	Target scheme reference
lgd_code	INTEGER ▾	Not Null ▾	FK → dim_lgd_geography.lg... ▾	Administrative boundary
reporting_month	DATE ▾	Not Null (Normalized to 1st o... ▾	None ▾	Monthly time-series anchor
target_units	INTEGER ▾	Not Null, Default 0 ▾	None ▾	Physical target (e.g., houses, tap lines)
achieved_units	INTEGER ▾	Not Null, Default 0 ▾	None ▾	Physically verified units completed
beneficiaries_total	INTEGER ▾	Not Null, Default 0 ▾	None ▾	Total unique beneficiaries served
beneficiaries_women	INTEGER ▾	Default 0 ▾	None ▾	Sub-metric count: Female beneficiaries
beneficiaries_sc_st	INTEGER ▾	Default 0 ▾	None ▾	Sub-metric count: SC/ST beneficiaries

Table 6: fact_detected_anomalies

Stores machine-detected divergence, gaps, and underperformance alerts.

Field Name	Type	Constraints	Relations / Foreign Keys	Description
anomaly_id	UUID ▾	PK, Default gen_random_uui... ▾	None ▾	Primary Key
scheme_id	UUID ▾	Nullable (Null if cross-sche... ▾	FK → dim_schemes.scheme... ▾	Scheme involved
lgd_code	INTEGER ▾	Not Null ▾	FK → dim_lgd_geography.lg... ▾	Location of anomaly
anomaly_type	VARCHAR(32) ▾	Check (anomaly_type IN ('U... ▾	None ▾	Categorical label
severity	VARCHAR(16) ▾	Check (severity IN ('LOW', '... ▾	None ▾	Operational escalation level
z_score	NUMERIC(5,2) ▾	Nullable ▾	None ▾	Statistical standard deviations from mean

Field Name	Type	Constraints	Relations / Foreign Keys	Description
evidence_payload	JSONB ▾	Not Null ▾	None ▾	Data points & formula variables driving alert
status	VARCHAR(16) ▾	Default 'OPEN' Check (statu... ▾	None ▾	Action resolution lifecycle
created_at	TIMESTAMPZ ▾	Default clock_timestamp() ▾	None ▾	Ingestion timestamp

2. Security & Access Rules (PostgreSQL Row-Level Security)

```
-- 1. Enable RLS on core fact tables

ALTER TABLE fact_scheme_allocations ENABLE ROW LEVEL SECURITY;

ALTER TABLE fact_scheme_beneficiaries_agg ENABLE ROW LEVEL SECURITY;

ALTER TABLE fact_detected_anomalies ENABLE ROW LEVEL SECURITY;

-- 2. Policy: Super Admins & Central Executives have universal read access

CREATE POLICY p_central_read_all_allocations

ON fact_scheme_allocations

FOR SELECT

TO authenticated

USING (

    (auth.jwt() ->> 'role') IN ('super_admin', 'central_executive', 'auditor')

);

-- 3. Policy: Department Nodal Officers can only read data related to their assigned Ministry

CREATE POLICY p_dept_nodal_read_own_scheme_allocations

ON fact_scheme_allocations

FOR SELECT

TO authenticated

USING (
```

```

(auth.jwt() →> 'role') = 'dept_nodal'

AND scheme_id IN (

    SELECT s.scheme_id

    FROM dim_schemes s

    WHERE s.ministry_id = (auth.jwt() →> 'assigned_ministry_id')::UUID

)

);

```

-- 4. Policy: District Magistrates can only view records within their assigned District LGD hierarchy

```

CREATE POLICY p_dm_read_local_geography

ON fact_scheme_allocations

FOR SELECT

TO authenticated

USING (

    (auth.jwt() →> 'role') = 'district_magistrate'

    AND lgd_code IN (

        SELECT geo.lgd_code

        FROM dim_lgd_geography geo

        WHERE geo.lgd_code = (auth.jwt() →> 'assigned_lgd_code')::INTEGER

        OR geo.parent_lgd = (auth.jwt() →> 'assigned_lgd_code')::INTEGER

    )

);

```

-- 5. Policy: Anomaly updates (acknowledgement/resolution) strictly limited to DMs in their district or Super Admins

```

CREATE POLICY p_update_anomalies_scope

```

ON fact_detected_anomalies

FOR UPDATE

TO authenticated

USING (

(auth.jwt() →> 'role') = 'super_admin'

OR (

(auth.jwt() →> 'role') = 'district_magistrate'

AND lgd_code = (auth.jwt() →> 'assigned_lgd_code')::INTEGER

)

)

WITH CHECK (

status IN ('ACKNOWLEDGED', 'RESOLVED')

);